



TRAINING PROGRAMME

AWS AI Practitioner & Generative AI

Use AI, machine learning and generative AI services on AWS responsibly

33 HOURS

FOR IT professionals, developers, analysts and managers working with AI on AWS

PREPARES FOR AWS Certified AI Practitioner (AIF-C01)



Mohamed Aziz BEN HAHA

mohamedazizbenhaha.netlify.app

Overview

This programme gives participants a practical understanding of AI, machine learning and generative AI on Amazon Web Services and prepares the AWS Certified AI Practitioner exam (AIF-C01). They learn the ML lifecycle and the ready-made AI services for vision, language, documents and speech, then the concepts behind foundation models, prompt engineering, and Amazon Bedrock with knowledge bases, agents and customisation. They use SageMaker Canvas and JumpStart, see what Amazon Q offers, and apply responsible AI, security and governance practices. Exam practice closes the content, and the programme ends with a final project: a generative AI assistant on Bedrock built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	AI & ML on AWS	AI and ML concepts, the ML lifecycle, ready-made AI services	6 h
02	Generative AI & Amazon Bedrock	Foundation models, prompting, Bedrock knowledge bases, agents and customisation	6 h
03	Tools, Responsible AI & Exam Preparation	SageMaker Canvas and JumpStart, Amazon Q, responsible AI, security, AIF-C01	9 h
04	Final Project	A generative AI assistant on Bedrock, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		33 h

Key facts

AUDIENCE	IT professionals, developers, analysts and managers working with AI on AWS
PREREQUISITES	None in AI. Familiarity with core AWS services helps (course: AWS Cloud Practitioner) but is not required
DURATION	33 h: 27 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs in the AWS console; exam practice
LAB ENVIRONMENT	One laptop per participant with a browser; one AWS sandbox account per team with a budget alert, provided by the organiser; Knowledge Bases on S3 Vectors; endpoints and vector stores deleted and Canvas logged out after each lab
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can choose the right AWS AI service for a use case, build a guarded RAG assistant with Bedrock, estimate its cost and sit the AIF-C01 exam

PHASE 01

6 H

AI & ML on AWS

Participants learn the vocabulary of AI and machine learning and the lifecycle of an ML project on AWS, then use ready-made services for images, text, documents, speech and translation without training a model. Validated lab: a document processed by Textract, its entities and sentiment detected by Comprehend, and the result read aloud by Polly.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	AI & ML Fundamentals	AI, ML and deep learning; supervised, unsupervised and reinforcement learning; the ML lifecycle on AWS; model metrics (accuracy, F1, AUC) and business metrics	Lecture + Workshop	3 h
1.2	AWS AI Services	Rekognition, Comprehend, Textract, Lex, Polly, Transcribe, Translate	Lecture + Lab	3 h

PHASE 02

6 H

Generative AI & Amazon Bedrock

Participants learn how foundation models work and where they fail, write effective prompts, then build with Amazon Bedrock: model choice in the playgrounds, retrieval-augmented generation with Knowledge Bases, Agents and model customisation. Validated lab: a Bedrock Knowledge Base that answers questions over a set of company documents with cited sources.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Generative AI Concepts	Foundation models, tokens, embeddings, use cases and limits	Lecture + Workshop	1.5 h
2.2	Prompt Engineering	Techniques and pitfalls	Lecture + Lab	1.5 h
2.3	Amazon Bedrock	Models and playgrounds, inference parameters (temperature, top-p), Knowledge Bases (RAG) and vector stores on AWS, Agents, customisation (fine-tuning, continued pre-training, distillation), model evaluation (ROUGE, BLEU, BERTScore, human review)	Lecture + Lab	3 h

PHASE 03

9 H

Tools, Responsible AI & Exam Preparation

Participants build a model without code in SageMaker Canvas and deploy a pretrained one from JumpStart, see the use cases of Amazon Q Business and Q Developer, apply responsible AI with Bedrock Guardrails and SageMaker Clarify, and secure and govern AI workloads before practising AIF-C01 questions. Validated lab: a Bedrock Guardrail that blocks a denied topic and masks personal data in the assistant's answers.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	SageMaker AI for Practitioners	SageMaker Canvas, JumpStart	Lecture + Lab	1.5 h
3.2	Amazon Q	Amazon Q Business and Amazon Q Developer use cases (exam scope); Amazon Quick (formerly Quick Suite) as the newer agentic workspace	Lecture + Workshop	1.5 h
3.3	Responsible AI	Bias, transparency, Bedrock Guardrails, SageMaker Clarify, Amazon A2I human review, SageMaker Model Cards	Lecture + Lab	1.5 h
3.4	Security & Governance	Shared responsibility, IAM, encryption, Macie, PrivateLink, CloudTrail, AWS Config, Audit Manager, AWS Artifact, data lineage	Lecture + Workshop	1.5 h
3.5	AIF-C01 Preparation	Practice questions and review	Lecture + Workshop	3 h

PHASE 04

6 H

Final Project

Teams of two or three build an assistant for a company's documents from a written brief, reusing the phase labs, then present it with its cost estimate as they would to a business sponsor.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, build the Knowledge Base over the company documents, connect one AI service such as Textract	Project	3 h
4.2	Guard & Estimate	Guardrails, test questions, cost estimate, architecture diagram	Project	1.5 h
4.3	Demo & Review	Live demo including a blocked request, cleanup; questions and feedback	Project	1.5 h

FINAL PROJECT

A Generative AI Assistant on Bedrock

An assistant that answers questions over company documents through a Bedrock Knowledge Base, reads scanned documents with an AI service such as Textract, is protected by Guardrails and comes with a cost estimate.

STACK AWS: Amazon Bedrock (Knowledge Bases, Guardrails), S3, Textract, AWS Pricing Calculator

DELIVERABLE Working assistant, test questions and results, cost estimate, live demo